AUTOMATED CREDIT CARD FRAUD DETECTION

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In this project, I developed a **machine learning-based system** to detect fraudulent credit card transactions using Python. Unlike conventional rule-based approaches, the program analyzes transactional patterns using statistical and machine learning techniques to identify anomalies indicative of fraud.

I worked with a publicly available credit card transactions dataset, performed exploratory data analysis, handled class imbalance, and applied multiple outlier detection models including **Isolation Forest**, **Local Outlier Factor**, and **One-Class SVM** to classify transactions as normal or fraudulent.

This project gave me **hands-on experience** with anomaly detection, supervised and unsupervised learning concepts, data preprocessing, and evaluation of machine learning models in cybersecurity contexts.

OBJECTIVE:

The main goals of this project were to:

1) Detect fraudulent credit card transactions using **machine learning models**.
2) Understand anomaly detection techniques like **Isolation Forest**, **Local Outlier Factor**, and **One-Class SVM**.
3) Perform exploratory data analysis **to understand class distribution & feature correlations**.
4) Gain **practical experience in preprocessing**, **model training**, **& evaluation** for **real-world cyber applications**.

KEY CONCEPTS LEARNED:

DATA EXPLORATION & VISUALIZATION:

- Loaded the dataset using **pandas** & inspected its structure.
- Checked for missing values; dataset is highly imbalanced with few fraudulent transactions.
- Visualized distribution of transaction amounts for fraud & normal case using histograms & log-scaled axes.
- Generated a heatmap of feature correlations to understand relationships among features.

SAMPLING & FEATURE SELECTION:

- Reduced dataset size for faster experimentation using random sampling **(frac=0.1)**.
- Separated the target column **(Class)** from input features for modeling.
- Converted data into **feature matrix X & target vector Y**.

ANOMALY DETECTION MODELS:

- **Isolation Forest** identifies anomalies by isolating observations in a random forest structure.
- **Local Outlier Factor (LOF)** detects outliers based on local density deviations of a data point with respect to its neighbors.
- **One-Class SVM** learns the boundary of normal transactions & flags deviations as fraud.

MODEL TRAINING & PREDICTION:

- Configured models with appropriate hyperparameters & contamination rates **(fraud-to-normal ratio)**.
- Trained each model on the sampled dataset.
- Converted model predictions to consistent format: **0 = Normal**, **1 = Fraud**.
- Calculated **errors**, **accuracy**, **& classification reports for evaluation**.

EVALUATION:

- **Isolation Forest:** **Accuracy ≈ 99.78%**, able to detect some fraudulent transactions.
- **Local Outlier Factor:** **Accuracy ≈ 99.68%**, lower recall on fraud detection due to dataset imbalance.
- **One-Class SVM:** Performance depended on **kernel & hyperparameter selection**.
- Evaluated models using **precision**, **recall**, **f1-score**, **& support metrics**.

DATA HANDLING:

- Used **numpy & pandas** for **numerical & tabular data handling**.
- Applied **matplotlib & seaborn** for **plotting distributions & correlations**.
- Ensured **reproducibility with a fixed random seed**.

PYTHON IMPLEMENTATION:

- Imported essentials libraries **(numpy, pandas, sklearn, matplotlib, seaborn)**.
- Loaded dataset with **pd.read_csv()**.
- Checked dataset info **(data.info())** and missing values.
- Split the dataset into fraud & normal transactions for analysis.
- Calculated the outlier fraction to parameterize anomaly detection models.
- Defined models in a dictionary & iterated to train & evaluate each one.
- Used loops to handle predictions & computed performance metrics.
- **Visualized class distributions, transaction amounts, and correlation matrices**.

PRACTICAL USE:

- **Real-Time Fraud Detection:** Demonstrates how machine learning models, such as Isolation Forest, Local Outlier Factor, and One-Class SVM, can detect fraudulent transactions as they occur, helping banks & financial institutions prevent monetary losses in real-time.
- **Handling Imbalanced Data:** Highlights the challenges of extremely imbalanced datasets, common in cybersecurity & financial fraud, and shows how sampling, outlier detection, and model tuning can mitigate these issues.
- **Risk Management & Decision Support:** Provides insight into deploying outlier detection models to flag suspicious transactions for further investigation, supporting operational risk management & automated alert systems.
- **Data-Driven Cybersecurity Analytics:** Strengthened skills in data preprocessing, exploratory data analysis, visualization, and Python programming to extract actionable insights from large-scale transactional datasets, enhancing practical cybersecurity & fraud detection capabilities.
- **Scalable Methodology:** Illustrates a framework that can be scaled to handle millions of transactions, making it relevant for real-world applications in banking, fintech, and payment processing systems.

CONCLUSION:

Exploring **large-scale transactional data** **&** **detecting fraudulent patterns** gave me valuable practical exposure…

I

Build a credit card fraud detection system using Python & machine learning.
Gained **hands-on experience** in anomaly detection & managing imbalanced datasets.
Learned to **preprocess**, **visualize**, **&** **analyze large financial datasets**.
Strengthened understanding of **model evaluation metrics** **&** **cyber applications**.
Improved Python & machine learning skills for **real-world data security scenarios**.